Skills Summary

Solving Quadratics Without Losing a Root or a Sign

Use this reference while completing your worksheet or when studying.

Skill 1 — Every square root gets a plus-or-minus

Rule: if $u^2 = k$ and $k > 0$, then $u = \pm\sqrt{k}$ — two numbers share one square, so one square root step produces *two* branches.

- Never divide both sides by something containing x: that erases the root where it equals zero. Factor instead.
- How many roots to expect: discriminant positive $\to$ two, zero $\to$ one (a double root), negative $\to$ none real.
- Finish by substituting every root back into the original equation.

Example: Solve $x^2 = 36$.
Step 1. The variable appears only as a square, so this is a square-root problem — and the square-root step is exactly where a root goes missing.
Step 2. $x = \pm\sqrt{36}$, which splits into the two branches $x = +6$ and $x = -6$.
$\boldsymbol{x = -6}$ or $\boldsymbol{x = 6}$

Try it: Solve $x^2 = 81$. Give the complete solution set.

check: $\boldsymbol{x = -9}$ or $\boldsymbol{x = 9}$

Skill 2 — Substitute signs, not just digits

Formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, $\quad \Delta = b^2 - 4ac$.

Two sign traps. $-b$ means *the opposite of* b: if b is negative, $-b$ is positive. And $-4ac$ becomes an *addition* whenever a and c have opposite signs. Write a, b, c with their signs in a list before substituting, and put every negative in brackets.

Example: Solve $x^2 - 6x + 5 = 0$ with the formula; give the larger root.
Step 1. Nothing is squared alone here, so the formula is the general tool — and b is negative, which is where the sign slip happens.
Step 2. $a = 1$, $b = -6$, $c = 5$, so $-b = 6$ and $\sqrt{b^2 - 4ac} = 4$: $x = \dfrac{6+4}{2}$.
The larger root is $\boldsymbol{x = 5}$.

Try it: Solve $x^2 - 2x - 8 = 0$ with the formula and give the larger root.

check: $\boldsymbol{x = 4}$

Skill 3 — Reading a factored equation correctly

Zero-product property: if $PQ = 0$ then $P = 0$ or $Q = 0$. Set *each factor* equal to zero and solve it as its own little equation.

- The root has the *opposite* sign to the constant in its factor.
- A factor with a coefficient needs two moves, not one: subtract, then divide.
- Count your roots: one factor, one root.

Example: Solve $(3x-2)(x+5)=0$.
Step 1. The product is already zero and already factored, so the zero-product property finishes it — no expanding.
Step 2. $3x-2=0$ needs two moves: $3x=2$, then divide by 3. And $x+5=0$ gives $x=-5$.
$\boldsymbol{x = \frac{2}{3}}$ or $\boldsymbol{x = -5}$

Try it: Solve $(2x-5)(x+1)=0$.

check: $\boldsymbol{x = \frac{5}{2}}$ or $\boldsymbol{x = -1}$

(!) Watch out: A solution set with one root is a claim, not a default. Either the discriminant is zero, or you lost a root at a square root or a division. Check which before you hand it in.